

Practice Problem Set 2

1. Let $X_1, \dots, X_n \stackrel{i.i.d.}{\sim} \text{Pareto}(\xi, \alpha_0)$ distribution, with pdf

$$f_X(x) = \frac{\alpha_0 \xi^{\alpha_0}}{x^{\alpha_0+1}}, \quad x > \xi > 0,$$

where $\alpha_0 > 1$ is known.

- (i) Find the MLE of ξ , say $\hat{\xi}_{\text{MLE}}$.
 - (ii) Find the distribution and expectation of $\hat{\xi}_{\text{MLE}}$.
 - (iii) Given that $\hat{\xi}_{\text{MLE}}$ is a complete-sufficient statistic for ξ , find the UMVUE of ξ .
2. Let $X_1, \dots, X_n$ be a random sample from **Geometric**(θ) distribution. Define $g(\theta) = P(X_1 \geq 4)$.
- (i) Find an unbiased estimator of $g(\theta)$.
 - (ii) Find an MLE of $g(\theta)$.
3. Let $X_1, \dots, X_n$ be i.i.d. random samples from a continuous distribution with pdf $f(x; \lambda)$, and CDF $F_X(x; \lambda)$.
- (i) Show that $U_i = -\log(F_X(X_i; \lambda)) \sim \text{Exponential}(1)$, for each $i = 1, \dots, n$.
 - (ii) Find a pivot based on $U_1, \dots, U_n$.
 - (iii) Using (ii) or otherwise, can you find a confidence interval for λ with confidence coefficient $(1 - \gamma)$, $\gamma \in (0, 1)$, when $f(x; \lambda) = \lambda x^{\lambda-1}$ for $x > 0$ and $\lambda > 0$.
4. Let X be the number of successes in n independent trials with probability $p \in (0, 1)$ of success, and let $\phi(x)$ be the UMP test, as in Q.12. (Problem Set 6), for testing $H_0 : p \leq p_0$ against $H_0 : p > p_0$ at level of significance $\alpha \in (0, 1)$.
- (i) For $n = 6$, $p_0 = 0.25$, and the levels $\alpha = 0.05, 0.1, 0.2$, determine the value of c and γ , and find the power of the test against $p_1 = 0.3, 0.4, 0.5, 0.6, 0.7$.
 - (ii) If $p_0 = 0.2$ and $\alpha = 0.05$, and it is desired to have power ≥ 0.9 against $p_1 = 0.4$, determine the necessary sample size (a) by using tables of the binomial distribution, (b) by using the normal approximation.
 - (iii) Use the normal approximation to determine the sample size required when $\alpha = 0.05$, power = 0.9, $p_0 = 0.01$, $p_1 = 0.02$.
 - (vi) Suppose $p_0 = 0.2$ and $n = 10$, and it is observed that $\sum_{i=1}^{10} X_i = 8$. Then find the p -value of the observation. What is the minimum level- α of significance which leads to rejection of this sample?

5. Let $X_1, \dots, X_n$ be a random sample from the density

$$f(x; a, b) = \frac{1}{b} e^{-(x-a)/b} I(x \geq a),$$

where $a \in \mathbb{R}$ and $b > 0$.

- (i) Determine the UMP level- α test for testing $H_0 : a = a_0$ against $H_1 : a = a_1 (< a_0)$, when b is assumed known.

- (ii) Show that the power of the test (i) is given by $\beta^*(a_1) = 1 - (1 - \alpha) \exp\{-n(a_0 - a_1)/b\}$.
- (iii) Find a UMP level- α test of $H_0 : a \geq a_0$ against $H_1 : a < a_0$, when b is unknown.
6. An auto manufacturer gives a bumper-to-bumper warranty for 3 years or 36,000 miles for its new vehicles. In a random sample of 60 of its vehicles, 20 of them needed five or more major warranty repairs within the warranty period. Estimate the true proportion of vehicles from this manufacturer that need five or more major repairs during the warranty period, with confidence coefficient 0.95. Interpret your findings. [Hint: Use CLT]
7. Let $U_1, \dots, U_n$ be a random sample from a Poisson distribution with parameter λ . Using CLT, derive a large sample confidence interval for λ with confidence coefficient $(1 - \alpha)$.
8. Let $X_1, \dots, X_n$ be a random sample from a $\text{Uniform}(\theta, \theta + 1)$ distribution. Find a 99% confidence interval for θ , using an appropriate pivot.
9. Let $Y_1, \dots, Y_n$ be a random sample from a $\text{Normal}(\mu, \sigma^2)$ distribution. A $100(1 - \alpha)\%$ confidence interval for σ^2 is given by $[(n - 1)S^2/U, (n - 1)S^2/L]$, where S^2 is the sample variance with divisor $(n - 1)$, and L and U are positive constants. Find L and U .
10. Let X be a random variable supported on $[-5, 5]$. Consider the testing problem $H_0 : X \sim f_0$ against $H_1 : X \sim f_1$, where

$$f_0(x) = \begin{cases} 1/10 & \text{if } x \in [-5, 5], \\ 0, & \text{otherwise,} \end{cases} \quad \text{and} \quad f_1(x) = \begin{cases} (5 - |x|)/25 & \text{if } x \in [-5, 5], \\ 0, & \text{otherwise.} \end{cases}$$

- (i) Using the Neyman-Pearson's lemma, find the MP level-0.05 test for testing H_0 against H_1 .
- (ii) Find the power of the test obtained in part (i). Is the test unbiased?
11. Suppose that $\mathbb{P}(X = 1) = \mathbb{P}(X = -1) = 1/2$. Define
- $$X_n = \begin{cases} -1 & \text{with probability } (n - 1)/2n \\ 1 & \text{with probability } (n - 1)/2n \\ e^n & \text{with probability } 1/n \end{cases}$$
- (i) Does X_n converge to X in probability?
- (ii) Does X_n converge to X in distribution?
- (iii) Does X_n converge to X in L^2 ?
12. Suppose that $X_n \sim \text{Normal}(0, 1/n)$ and let X be a random variable with distribution $F_X(x) = 0$ if $x < 0$ and $F_X(x) = 1$ if $x \geq 0$.

- (i) Does X_n converge to X in probability? (Prove or disprove).
- (ii) Does X_n converge to X in distribution? (Prove or disprove).

13. For each $n = 1, 2, 3, \dots$, suppose that X_n is a discrete random variable with range $\{\frac{1}{n}, \frac{2}{n}, \dots, 1\}$ and

$$\mathbb{P}\left(X_n = \frac{j}{n}\right) = \frac{2j}{n(n+1)}, \quad j = 1, 2, \dots, n.$$

Does $X_1, X_2, \dots$ converge in distribution to some random variable X ? If so, find the distribution of X .